

MINA BH ARSANIOUS

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EDUCATION

Bachelor of Science in Pure Mathematics

2024 - Present

Cairo University, Giza

A motivated sophomore at Cairo University majoring in pure mathematics.

High School Diploma

2021 - 2024

Assiut STEM School, Assyut

Ranked the best Egyptian secondary school in Upper Egypt, Assiut STEM School is the product of a partnership between the U.S. Agency for International Development (USAID) and its sub-recipients the ECASE project to provide technical assistance to the Ministry of Education to establish a model STEM school in each Egyptian governorate. Assiut STEM school was founded in 2015 and is a two semester based school that runs from September to June. Accreditation: The school is accredited by the United States Agency for International Development (USAID) and the Egyptian Ministry of Education.

SKILLS AND INTERESTS

Interests	Foundations of Mathematics & Mathematical Philosophy, Combinatorics, Formalization and Proof Assistants, Mathematical Visualization, Algebra
Skills	LaTeX, Desmos, Technical Writing, Microsoft Excel, General Coding

EXPERIENCES

International Research Scholar

Feb 2023 - Aug 2023

Pioneer Academics, Remotely

I had the opportunity to conduct individual research on Fibonacci numbers and visual proofs under the guidance of Professor Gregory Dresden who is Professor of Mathematics at Washington and Lee University. Our work began with a deep dive into “Proofs That Really Count” by Art Benjamin and Jennifer Quinn, where we explored tiling problems using squares and dominos. From there, I developed original ideas and, together with Professor Dresden, produced and proved new mathematical formulas and theorems. One unique aspect of my research involved translating chemical properties, such as carbon bonding rules in complex hydrocarbons, into mathematical sequences. This approach allowed me to derive new identities, including one that aligns with sequence A006356 in the Online Encyclopedia of Integer Sequences (OEIS). My formula for the partial sums of this sequence is now featured on the OEIS webpage. Additionally, I worked on identities involving specific digit sequences (0, 1, 2, and 3) that avoid certain patterns, which resulted in another entry on OEIS (A214997). In our research group discussions, I contributed actively, occasionally catching technical errors and engaging in our shared problem-solving. The experience enhanced my understanding of both collaborative and independent research.

Speaker & Student Supervisor

2024 - Present

Math Club, Cairo University

I delivered two departmental talks: “The Math of Molecules” (first year), presenting the mathematics behind my Pioneer Academics research on combinatorial techniques and chemically inspired sequences; and “Is Mathematics Real?” (second-year, first term), an expository talk on mathematical objects and their ontological status. At the start of the current semester I began supervising the club with fellow students, leading recruitment efforts for incoming students and developing a program of seminars and activities to engage newcomers and build a community of math enthusiasts (planning stage).

ACHIEVEMENTS

11th Nationally American Math Olympiad 2021	Oct 2021
Recipient of Kangaroo Math Excellence Award 2021	Feb 2022
Quarter-Finalist in Climate Science Olympiad 2022	May 2022
Certificate of Contribution to Pioneer Co-Curricular Summit 2022	Jul 2022
Recognized Participant in NASA Space Apps 2022	Sep 2022
Certificate of Recognition for Lunar Settlement Design at EGSDCTM 2022	Sep 2022

POSITION OF RESPONSIBILITY

STEM Assuit Community Team Leader <i>Assiut STEM School</i>	Sep 2022 - July 2023
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Alongside my academic pursuits, I have been actively involved in a range of non-profit initiatives. As the STEM Assuit Community Team Leader, I have been dedicated to raising the academic level of my colleagues through a variety of activities.

EXTRA-CIRRICULAR

CS Tutor <i>My Church Center</i>	Jun 2023 - Aug 2023
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I have served as a CS teacher in the church center in our village, devoting 10 hours a week during the summer holidays to teaching courses in Photoshop, PowerPoint, and SketchUp. My students went on to win first place in the Saint Mark Festival competition in CS, which is highly competitive as it involves participants from all around Egypt.

Chess Club Trainer <i>My Church Center</i>
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In this role, I train people in chess at my local church center. I am passionate about helping others develop their skills and enjoy sharing my knowledge with others.

DECLARATION

I hereby declare that all the details furnished above are true to the best of my knowledge and belief.